We create chemistry

Safety data sheet

1. Identification

Product identifier

Status Milled

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, herbicide

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Eye Dam./Irrit. 2B
Skin Sens. 1
Aquatic Acute 1
Aquatic Chronic 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

H317	May cause an allergic skin reaction.
H320	Causes eye irritation.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read label before use.

Precautionary Statements (Prevention):

P261	Avoid breathing dust.
P264	Wash contaminated body parts thoroughly after handling.
P272	Contaminated work clothing should not be allowed out of the workplace.
P280	Wear protective gloves/clothing/eye protection.

Precautionary Statements (Response):

P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P333 + P311	If skin irritation or rash occurs: Call a POISON CENTER or doctor/physician.
P391	Collect spillage.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P337 + P311	If eye irritation persists: Call a POISON CENTER or doctor/physician.

Precautionary Statements (Disposal):

P501	Dispose of contents/container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: Sodium alkylnaphthalenesulfonate blend, Isoxadifen-ethyl, Residues (petroleum), catalytic reformer fractionator, sulfonated,polymers with formaldehyde, sodium salts, sodium diflufenzopyr

Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

crop protection product, herbicide, water dispersible granules

Contains: sodium salt, dicamba (ISO)

Hazardous ingredients (GHS)
According to UN GHS criteria

sodium 3,6-dichloro-o-anisate

Content (W/W): 44 %	Aquatic Acute 2
CAS Number: 1982-69-0	Aquatic Chronic 2
EC-Number: 217-846-3	H401, H411
INDEX-Number: 607-243-00-7	

sodium diflufenzopyr

Content (W/W): 17,11 %	Acute Tox. 5 (oral)
CAS Number: 109293-98-3	Aquatic Acute 1
	Aquatic Chronic 1
	H303, H400, H410

Isoxadifen-ethyl

Content (W/W): 10,04 % CAS Number: 163520-33-0	Acute Tox. 4 (oral) Skin Sens. 1 Aquatic Acute 1 Aquatic Chronic 1 H302, H317, H400, H410

Sodium alkylnaphthalenesulfonate blend

Content (W/W): < 10 %	Acute Tox. 4 (oral) Skin Corr./Irrit. 2 Eye Dam./Irrit. 2A Skin Sens. 1 Aquatic Acute 3 Aquatic Chronic 3 H319, H315, H302, H317, H402, H412

Residues (petroleum), catalytic reformer fractionator, sulfonated,polymers with formaldehyde, sodium salts

Content (W/W): < 10 % CAS Number: 68425-94-5	Eye Dam./Irrit. 2A H319

methanol

Content (W/W): < 3 % CAS Number: 67-56-1 EC-Number: 200-659-6 INDEX-Number: 603-001-00-X	Flam. Liq. 2 Acute Tox. 3 (Inhalation - vapour) Acute Tox. 3 (oral) Acute Tox. 3 (dermal) STOT SE (Central nervous system, Optic nerve) 1 H225, H311, H331, H301, H370 <u>Specific concentration limit:</u> STOT SE 2: 3 - < 10 % STOT SE 1: >= 10 %

Silicon dioxide

Content (W/W): < 20 % CAS Number: 61790-53-2	

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed
Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11., Further symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media
Suitable extinguishing media:
foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:
carbon dioxide

Special hazards arising from the substance or mixture
carbon monoxide, Carbon dioxide, nitrogen oxides, halogenated compounds
The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters
Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Avoid dust formation. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into drains/surface waters/groundwater. Do not discharge into the subsoil/soil.

Methods and material for containment and cleaning up

For small amounts: Contain with dust binding material and dispose of.
For large amounts: Sweep/shovel up.
Avoid raising dust. Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations.

7. Handling and Storage

Precautions for safe handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Avoid dust formation. Dust can form an explosive mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect against moisture. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

67-56-1: methanol
61790-53-2: Silicon dioxide

Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:
Suitable chemical resistant safety gloves (EN 374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	solid, granules
Colour:	light brown
Odour:	characteristic
Odour threshold:	Not determined due to potential health hazard by inhalation.
pH value:	approx. 6,5 - 8,5 (1 %(m), 25 °C)
Melting point:	The product has not been tested.
Boiling point:	The product has not been tested.
Flash point:	not applicable, the product is a solid
Evaporation rate:	not applicable
Flammability:	Product is combustible.

Lower explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.

Upper explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.

Vapour pressure:
The product has not been tested.

Relative vapour density (air):
not applicable

Solubility in water: dispersible

Information on: sodium 3,6-dichloro-o-anisate
Partitioning coefficient n-octanol/water (log Kow): -1,88
(pH value: 6,8)
Information on: sodium diflufenzopyr
Partitioning coefficient n-octanol/water (log Kow): 0,037
(20 °C; pH value: 7)
Information on: Isoxadifen-ethyl
Partitioning coefficient n-octanol/water (log Kow): 3,8

Self ignition: not determined

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.

Viscosity, dynamic:
not applicable, the product is a solid

Explosion hazard: not explosive

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Bulk density: 0,717 kg/l
(25 °C)

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid
See MSDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong acids, strong oxidizing agents, strong bases

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Experimental/calculated data:
LD50 rat (oral): > 2.000 mg/kg
No mortality was observed.

LC50 rat (by inhalation): > 5,3 mg/l

LD50 rat (dermal): > 2.000 mg/kg
No mortality was observed.

Irritation

Assessment of irritating effects:
Not irritating to the skin. May cause slight irritation to the eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: Slightly irritating. (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Sensitization after skin contact possible.

Experimental/calculated data:
Buehler test guinea pig: sensitizing

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: methanol
Assessment of carcinogenicity:
In long-term studies in rats and mice in which the substance was given by inhalation, a carcinogenic effect was not observed. In long-term animal studies in which the substance was given in the drinking water in high concentrations, a carcinogenic effect was observed. These effects are not relevant to humans at occupational levels of exposure.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: methanol
Assessment of teratogenicity:
The results of animal studies gave indication of a developmental toxic/teratogenic effects with high doses.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: methanol
Assessment of repeated dose toxicity:
The substance may cause blindness after repeated ingestion. The substance may cause blindness after repeated inhalation.

Aspiration hazard

No aspiration hazard expected.
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: sodium 3,6-dichloro-o-anisate
Toxicity to fish:
LC50 (96 h) 135 mg/l, Oncorhynchus mykiss

Information on: sodium diflufenzopyr
Toxicity to fish:
LC50 (96 h) 106 mg/l, Oncorhynchus mykiss (Fish test acute, static)
The statement of the toxic effect relates to the analytically determined concentration.
Analogous: Assessment derived from products with similar chemical character.

LC50 (96 h) > 138 mg/l, Cyprinodon variegatus (Fish test acute, static)
The statement of the toxic effect relates to the analytically determined concentration.

Analogous: Assessment derived from products with similar chemical character.

Information on: Isoxadifen-ethyl
Toxicity to fish:
LC50 (96 h) 0,22 mg/l, Lepomis macrochirus (other)

LC50 (96 h) 0,34 mg/l, Oncorhynchus mykiss (other)

Information on: sodium 3,6-dichloro-o-anisate
Aquatic invertebrates:
EC50 (48 h) 110 mg/l, Daphnia magna

Information on: sodium diflufenzopyr
Aquatic invertebrates:
EC50 (48 h) 15 mg/l, Daphnia magna (Daphnia test acute, static)
The statement of the toxic effect relates to the analytically determined concentration.
Analogous: Assessment derived from products with similar chemical character.

LC50 (96 h) 19 mg/l, Mysidopsis bahia (OPP 72-3 (EPA-Guideline), static)
The details of the toxic effect relate to the nominal concentration.
Analogous: Assessment derived from products with similar chemical character.

Information on: Isoxadifen-ethyl
Aquatic invertebrates:
LC50 (48 h) > 0,51 mg/l, Daphnia magna (other)
The product has low solubility in the test medium. A saturated solution has been tested.

Information on: sodium 3,6-dichloro-o-anisate
Aquatic plants:
EC50 >= 41 mg/l, algae

Information on: sodium diflufenzopyr
Aquatic plants:
EC50 (5 d) 0,11 mg/l (growth rate), Selenastrum capricornutum (Algal growth inhibition test, static)
The statement of the toxic effect relates to the analytically determined concentration.

No observed effect concentration (5 d) 0,0078 mg/l (growth rate), Selenastrum capricornutum (Algal growth inhibition test, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Isoxadifen-ethyl
Aquatic plants:
EC50 (72 h) > 1,26 mg/l, Pseudokirchneriella subcapitata (other)
The product has low solubility in the test medium. A saturated solution has been tested.

Information on: sodium 3,6-dichloro-o-anisate
Chronic toxicity to fish:
No observed effect concentration (33 d) > 10 mg/l, Pimephales promelas

Information on: sodium diflufenzopyr
Chronic toxicity to fish:
No observed effect concentration (28 d) 5 mg/l, Pimephales promelas (OPP 72-4 (EPA-Guideline), semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: sodium diflufenzopyr
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 1,9 mg/l, Daphnia magna (Daphnia test chronic, semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: sodium 3,6-dichloro-o-anisate
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: sodium diflufenzopyr
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Isoxadifen-ethyl
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). The product has not been tested.

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: sodium 3,6-dichloro-o-anisate
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Information on: sodium diflufenzopyr
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Information on: Isoxadifen-ethyl
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is possible.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: sodium 3,6-dichloro-o-anisate
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.

Information on: sodium diflufenzopyr
Assessment transport between environmental compartments:
Volatility: The substance will slowly evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

UN number	UN3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains ISOXADIFEN-ETHYL)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

RID

UN number	UN3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains ISOXADIFEN-ETHYL)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Inland waterway transport

ADN

UN number	UN3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains ISOXADIFEN-ETHYL)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

UN number:	UN 3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains ISOXADIFEN-ETHYL)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	None known

Air transport

IATA/ICAO

UN number:	UN 3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains ISOXADIFEN-ETHYL)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

To avoid risks to man and the environment, comply with the instructions for use.

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Sens.	Skin sensitization
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Acute Tox.	Acute toxicity

Skin Corr./Irrit.	Skin corrosion/irritation
Flam. Liq.	Flammable liquids
STOT SE	Specific target organ toxicity — single exposure
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.
H303	May be harmful if swallowed.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H402	Harmful to aquatic life.
H412	Harmful to aquatic life with long lasting effects.
H225	Highly flammable liquid and vapour.
H311	Toxic in contact with skin.
H331	Toxic if inhaled.
H301	Toxic if swallowed.
H370	Causes damage to organs (Central nervous system, Optic nerve).

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.